

CREE LAKE, SK, Canada

WMO: 729200

Lat: 57.350N

Lon: 107.130W

Elev: 495

StdP: 95.52

Time Zone: -6.00 (W06)

Period: 86-96

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-40.7	-37.9	-40.2	0.1	-36.8	-38.3	0.1	-35.3	8.9	-9.3	7.8	-12.3	1.3	260	0.718

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	27.0	16.5	24.7	15.4	22.5	14.3	17.7	24.8	16.5	22.7	15.4	20.8	3.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.3	11.5	19.5	14.1	10.7	18.0	13.1	9.9	17.2	51.9	24.9	47.9	22.9	44.8	20.9	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.2	6.4	DB	-44.2	30.6	3.7	2.2	-46.9	32.1	-49.0	33.4	-51.0	34.7	-53.7	36.3	(4)
(5)			WB	-40.0	19.4	3.5	1.6	-42.5	20.6	-44.6	21.5	-46.5	22.5	-49.1	23.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.8	-20.6	-18.3	-11.1	-1.7	6.9	13.5	15.9	14.6	8.6	0.0	-12.1	-18.8	(6)	
	DBStd	14.94	9.21	9.65	8.68	6.19	5.17	3.74	3.17	4.16	4.31	6.27	7.44	10.17	(7)	
	HDD10.0	4838	947	793	653	352	127	13	0	7	78	312	664	893	(8)	
	HDD18.3	7401	1206	1027	911	602	357	149	92	132	293	568	914	1151	(9)	
	CDD10.0	517	0	0	0	1	30	117	183	148	36	3	0	0	(10)	
	CDD18.3	34	0	0	0	0	1	3	16	14	0	0	0	0	(11)	
	CDH23.3	342	0	0	0	0	17	37	124	161	3	0	0	0	(12)	
	CDH26.7	71	0	0	0	0	2	2	23	43	0	0	0	0	(13)	
	Wind	WSAvg	2.9	2.6	2.7	2.9	3.0	3.0	2.8	2.9	3.4	3.4	2.8	2.4	(14)	
(15) Precipitation	PrecAvg	476	19	16	17	23	30	66	93	70	51	39	25	23	(15)	
	PrecMax	626	35	39	43	59	68	142	163	111	112	86	62	39	(16)	
	PrecMin	351	6	2	4	5	3	20	44	24	17	11	8	8	(17)	
	PrecStd	77	7	9	9	12	19	34	38	25	29	19	16	9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.7	6.8	7.9	15.5	25.9	26.7	29.4	30.8	22.8	17.8	3.1	1.5	(19)	
		MCWB	1.2	2.6	2.7	7.5	13.8	15.1	17.7	19.6	14.5	10.3	1.3	0.2	(20)	
	2%	DB	-0.7	2.5	5.7	11.3	21.0	24.3	26.7	27.3	19.8	13.9	0.7	-1.0	(21)	
		MCWB	-1.6	0.2	1.6	5.4	12.0	13.8	16.9	17.1	12.5	8.3	-0.3	-1.9	(22)	
	5%	DB	-4.5	-0.9	3.5	8.8	18.3	22.4	24.2	24.8	17.8	10.6	-0.6	-3.3	(23)	
		MCWB	-5.1	-2.4	0.7	3.9	10.5	13.0	15.7	16.1	11.7	6.5	-1.2	-4.1	(24)	
	10%	DB	-7.6	-4.1	1.1	6.6	15.8	20.4	22.1	21.9	15.3	7.9	-2.4	-5.6	(25)	
		MCWB	-8.1	-5.0	-0.5	2.4	9.0	12.5	14.8	15.1	10.5	5.1	-3.0	-6.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.4	2.6	3.2	7.8	14.4	16.6	19.1	20.4	14.8	11.1	1.5	0.4	(27)	
		MCDB	3.6	6.0	7.0	14.0	23.4	21.8	24.4	28.8	21.5	17.1	2.6	1.5	(28)	
	2%	WB	-1.2	0.2	2.1	5.7	12.6	15.3	17.8	18.1	13.5	8.6	0.0	-1.8	(29)	
		MCDB	-0.2	2.3	5.1	10.6	20.0	20.5	24.6	25.3	18.3	13.0	0.6	-1.1	(30)	
	5%	WB	-4.9	-2.2	0.8	4.2	11.1	14.5	16.6	16.6	12.4	6.9	-1.3	-4.0	(31)	
		MCDB	-4.3	-0.8	3.2	8.3	17.4	19.8	22.9	23.1	16.6	10.5	-0.5	-3.1	(32)	
	10%	WB	-7.9	-4.8	-0.4	2.7	9.6	13.6	15.6	15.5	11.1	5.3	-2.9	-6.0	(33)	
		MCDB	-7.4	-3.9	1.4	6.1	15.1	19.0	20.7	21.2	14.8	7.8	-2.3	-5.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.6	11.4	12.7	10.6	11.0	10.0	9.2	9.6	8.3	6.8	7.3	8.9	(35)	
		MCDBR	12.8	11.6	12.9	12.6	14.5	13.2	12.1	13.9	11.8	11.3	7.8	8.9	(36)	
	5% WB	MCWBR	12.2	9.9	9.5	7.8	7.4	5.6	4.8	5.5	6.0	6.9	7.2	8.5	(37)	
		MCWBR	13.0	11.2	12.5	11.9	13.4	10.5	10.7	12.7	10.7	10.8	7.6	8.5	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.226	0.257	0.260	0.309	0.324	0.359	0.385	0.344	0.303	0.278	0.238	0.205	(40)	
		taud	2.202	2.236	2.314	2.331	2.475	2.395	2.358	2.456	2.595	2.560	2.287	2.152	(41)	
	Edn at Noon	Edn at Noon	675	804	912	907	915	880	845	863	861	795	656	593	(42)	
		Edh at Noon	48	74	90	106	99	109	111	94	70	55	44	37	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.67	1.67	3.37	4.99	5.62	5.78	5.38	4.30	2.83	1.46	0.59	0.42	(44)	
	RadStd	RadStd	0.05	0.10	0.14	0.29	0.63	0.46	0.34	0.34	0.35	0.21	0.07	0.05	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page